

Independence, Algebra, And Action

Compiling the Complex Scalar Field and Hamilton's Principle

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Abstract

Standard quantum mechanics assumes complex amplitudes. Classical and quantum mechanics assume Hamilton's principle. Neither is derived; both are imported. This paper compiles both from the same five primitive constraints $\{C_0, C_1, C_{1a}, C_2, C_5\}$ established in Paper 1. Two main theorems are proved. First, the Division Algebra Theorem: the unique admissible scalar continuation algebra is the complex numbers C , established by reducing to the Frobenius candidates $\{R, C, H\}$, eliminating R via C_5 's dual degree-of-freedom requirement, and eliminating H via the simplicity of H as a real algebra combined with C_{1a} 's independence constraint. Second, the Extremal Theorem: a field configuration is locally admissible under the compiled constraints if and only if it extremizes the action functional $\delta S = 0$. Hamilton's principle is the global expression of local C_1 -consistency. A compiled Measurement Theorem follows from the channel decomposition forced by C_5 : collapse is a C_2 history-update, not a C_1 transition. The independence constraint $[T_A, T_B] = 0$ serves as the structural spine: it compiles conservation (C_3 to L_3), forces the complex scalar field (Division Algebra Theorem), and will constrain gauge sector factorization. One axiom. Three compiled results.

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§1 Introduction

Two of the deepest assumptions in physics are that quantum amplitudes are complex-valued and that physical trajectories extremize the action. Both are usually treated as foundational inputs. The goal of this paper is to show both are compiled consequences of a primitive constraint set that mentions neither.

The companion paper (Paper 1) compiled: conservation, locality, the free-field Lagrangian L , and the wave equation, from the five primitive constraints $\{C_0, C_1, C_{1a}, C_2, C_5\}$. The present paper derives two further results from the same set, and establishes the structural connection between them.

The two theorems share the same spine. The independence constraint C_{1a} — that independent distinctions continue independently — has the algebraic form $[T_A, T_B] = 0$ for independent operators. This single condition eliminates quaternionic scalar algebras (Division Algebra Theorem), blocks coordinated creation-destruction channels (compiling conservation), and will factorize the gauge group (T_5, T_6) . Section 7 develops this as the Independence Spine Theorem.

Paper scope. Both theorems concern the free field (Paper 1, C_8). Interactions require C_9 (equivalence class coupling) and are addressed in Paper 4. The Measurement Theorem (Section 5) is fully compiled. The Born rule requires one additional input (C_7 geometry; see Section 5.3 and open problem Ω_B).

§2 The Primitive Constraints

The irreducible primitive set established in Paper 1 is $\{C_0, C_1, C_{1a}, C_2, C_5\}$. Conservation (L_3) and locality (L_4) are compiled consequences of this set, not primitives. The definitions are reproduced here for self-containment.

C_0 (Distinction exists): At least two states exist that are not identical. This is the minimum condition for any structure to be meaningful.

C_1 (Continuation exists): Every admissible state s admits at least one successor state s' with $s' \neq s$ under the transition algebra. Terminal states and stationary continuation are excluded. The identity element e of the composition algebra (empty composition, satisfying $e \circ T = T$) is required by the algebra but is not itself a physical continuation event (Lemma B, Section 4.2).

C1a (Independence Preservation): Independent distinctions remain independently continuable. The continuation of A cannot depend on the presence or continuation of B when A and B are independent.

C2 (History Preservation): Every continuation preserves the set of states reachable from prior states in the history. C2 forbids deletion of accessible states from the transition graph and establishes the ordering relation from reachability.

C5 (Observable Event Primitive): An observable event is the minimum continuation that simultaneously carries two independent degrees of freedom: propagation (a continuous spatial datum) and residue (a distinct temporal datum). C5 bridges transition structure and observation, and forces $\dim_R(D) \geq 2$ for any admissible scalar algebra D (Lemma D, Section 4.2).

§3 The Independence Constraint and Its Algebraic Form

C1a is the primitive that does the most structural work in this paper. Its algebraic form is derived immediately.

Let T_A and T_B be continuation operators acting on independent subsystems A and B. Their joint continuation must factor as $T_{AB} = T_A \otimes T_B$. Factorization requires that the result is independent of which update was applied first. Otherwise the joint continuation depends on an ordering between independent events, which is additional structure not licensed by Co to C2. Therefore:

$$[T_A, T_B] = T_A T_B - T_B T_A = 0 \quad \text{for all independent } A, B$$

Non-commutativity of independent continuation operators is a direct C1a violation by definition, not a violation of an algebraic nicety. This single equation is the structural spine of the paper.

Scope. This commutation requirement applies to the scalar continuation algebra — the algebra of weights on independent continuation amplitudes. Physical symmetry operators (angular momentum generators, Lorentz boosts, gauge generators) act at a different layer and are not required to commute. This restriction is explicit and essential throughout.

§4 The Division Algebra Theorem

§4.1 Frobenius Reduction [C]

Theorem [C]. *The admissible scalar continuation algebra D is a finite-dimensional division algebra over R .*

Proof. The compiled linear state space ($C_1 + C_{1a}$, Paper 1 §4.1) is a vector space over some scalar field D . Four properties of D follow from the primitives:

No zero divisors (from $C_0 + C_1$): C_0 requires nonzero states to remain distinguishable under scalar multiplication. If scalar $d \neq 0$ maps nonzero state ψ to zero, two distinct states become indistinguishable — a C_0 violation. Additionally, C_1 requires that every continuation produces a nonzero successor amplitude. If two nonzero amplitudes a and b satisfy $a \cdot b = 0$ in D , then applying amplitude a followed by b to a lawful continuation produces a terminated (zero) amplitude state — a C_1 violation. Therefore D has no zero divisors from both C_0 and C_1 .

Unit element (from C_1): C_1 requires the composition algebra to have an empty composition (Lemma B). This is the unit element of D .

Division algebra (from no zero divisors + unit + finite dimension): For finite-dimensional associative algebras over a field, the combination of no zero divisors and a unit element is equivalent to being a division algebra. Every nonzero element has a two-sided inverse.

Over R and finite-dimensional (from $C_5 + M_1$): Real scaling is the minimal requirement for continuous amplitude variation. C_5 forces $\dim_R(D) \geq 2$ (see Lemma D). The minimality principle (M_1) selects the minimal structure with no unconstrained degrees of freedom: $\dim_R(D) = \text{exactly } 2$. Infinite-dimensional scalar structures carry unconstrained degrees of freedom not licensed by any primitive and are excluded by M_1 .

Frobenius (1877): The only finite-dimensional division algebras over R are R (dimension 1), C (dimension 2), and H (dimension 4). Therefore $D \in \{R, C, H\}$. $\square$

§4.2 The Four Admissible Conditions

Any admissible scalar continuation algebra must satisfy four conditions, each compiled from the primitives. Together with Frobenius, these identify C as the unique survivor.

Lemma A (Associativity) [C]

A physical continuation history is an ordered sequence of lawful continuations. Any algebra representing that history must assign the same aggregate continuation map regardless of how the sequence is parenthesized. If associativity failed, the parenthesization — a bookkeeping convention — rather than the physical history would determine the outcome, violating Co. Therefore D is associative. $\square$

Lemma B (Identity) [C]

Sequential composition requires a neutral element — the empty composition e satisfying $e \circ T = T \circ e = T$ for all T . The empty composition is not a continuation event; it represents the absence of composition, a structural requirement of the algebra. C1 excludes stationary continuation ($T(s) = s$ as a physical event). It does not exclude the neutral element required by the composition algebra. These apply at different layers: e is required by the algebra and excluded as a physical continuation by C1. No contradiction. Identity is correctly placed in the representational layer. $\square$

Lemma C (Commutativity of the scalar algebra) [C]

The scalar weight on a continuation amplitude is itself a continuation operation $\psi \mapsto a \cdot \psi$. For independent amplitudes ψ_A and ψ_B with independent weights a_A and a_B , applying the weights in either order must give the same result — otherwise the combined amplitude depends on ordering between independent operations, which is a direct C1a violation. Therefore $a_A \cdot a_B = a_B \cdot a_A$ for all elements of D. D is commutative. $\square$

Lemma D (Two-dimensional) [C + M1]

C5 requires every observable event to simultaneously carry two independent degrees of freedom: propagation and residue. These are not the same degree of freedom. A one-dimensional algebra has only one real degree of freedom and its only intrinsic transformation generates Z_2 , which is discrete and cannot encode continuous propagation. Therefore:

C5 forces: $\dim_R(D) \geq 2$ (lower bound)

M1 selects: $\dim_R(D) = 2$ (minimality)

Higher-dimensional structures (quaternions, octonions, and beyond) carry additional degrees of freedom not licensed by any primitive. They are excluded by M1, not by a physical prohibition. The paper answers 'why not four?' with: four is not forced. Two is forced, and M1 selects the minimum forced structure. $\square$

§4.3 The Elimination

§4.3.1 Elimination of R [C from C₅]

Theorem [C]. $D = R$ is inadmissible.

Proof. An amplitude $a \in R$ has one independent real degree of freedom. The only intrinsic transformation on R -valued amplitudes is $a \mapsto -a$, generating $Z_2 = \{+1, -1\}$ — a discrete two-element group. C_5 requires continuous phase rotation: different propagation directions must be continuously interpolable. The group encoding this is $U(1) = \{e^{i\theta} : \theta \in [0, 2\pi)\}$, which is connected. Z_2 is not connected — no continuous path between $+1$ and -1 exists through group elements. Therefore $U(1)$ is not intrinsically realizable over R . Introducing the second real degree of freedom required to realize $U(1)$ is exactly passing from R to C . R fails Lemma D's lower bound. $\square$

§4.3.2 Elimination of H [C from C_{1a}]

Theorem [C]. $D = H$ is inadmissible.

Key fact: H is simple as a real algebra. Its only two-sided ideals are $\{0\}$ and H . Proof: H is a division algebra, so every nonzero element is invertible; any nonzero ideal contains an invertible element and therefore contains all of H .

Consequence for representations. Let $\rho : H \rightarrow \text{End}(V)$ be any R -algebra homomorphism. The kernel $\ker(\rho)$ is a two-sided ideal of H . By simplicity, $\ker(\rho)$ is either $\{0\}$ or H . Exactly two cases:

Case A — faithful ($\ker = \{0\}$): ρ is injective. In the scalar continuation interpretation, every nonzero element of D acts as an admissible amplitude scaling — that is, as a scalar continuation update applied to independent subsystems. The C_{1a} requirement applies to all such updates: independent scalar operations must commute when applied to independent factors. But the image $\rho(H)$ is isomorphic to H as a real algebra and inherits H 's non-commutativity. Take standard quaternion basis elements i and j : $\rho(i) \cdot \rho(j) = \rho(i \cdot j) = \rho(k)$, while $\rho(j) \cdot \rho(i) = \rho(j \cdot i) = \rho(-k) = -\rho(k)$. Therefore $[\rho(i), \rho(j)] = 2\rho(k) \neq 0$. Two independent phase updates on independent subsystems give different results depending on order. Direct C_{1a} violation.

Case B — trivial ($\ker = H$): ρ maps everything to zero. The scalar algebra has no quaternionic content and is not a representation of H as a nontrivial scalar algebra.

No Case C exists. Simplicity exhausts the possibilities. Every R -algebra homomorphism from H is either faithful (Case A, C_{1a} violated) or trivial (Case B, exits H). A global ordering convention to resolve non-commutativity introduces additional structure not licensed by Co—

C₅ — the same C_{1a} violation in different form. Therefore no representation of genuinely H-valued amplitudes satisfies C_{1a}. □

§4.3.3 Survival of C [C]

Theorem [C]. *C satisfies all four compiled conditions.*

Invertibility: $C \setminus \{0\}$ is a field — every nonzero element has a unique inverse. ✓

Continuous phase: $U(1) = \{e^{i\theta}\}$ acts intrinsically on C by multiplication. Connected, compact, one-dimensional. The unique group with these properties. ✓

Commutativity: $z_A \cdot z_B = z_B \cdot z_A$ for all $z_A, z_B \in C$. $[T_A, T_B] = 0$ is automatic. ✓

Norm-compatible distinguishability: $|z|^2 = z \cdot \bar{z} \geq 0$, equality only at $z = 0$. ✓

C is the unique element of $\{R, C, H\}$ satisfying all four conditions. □

§4.4 The U(1) Connection to Gauge Structure

The phase group of C is $U(1) = \{e^{i\theta} : \theta \in [0, 2\pi)\}$ — the minimal connected compact one-dimensional Lie group. This is identical to the first factor of the Standard Model gauge group $U(1) \times SU(2) \times SU(3)$.

C₅ requires two independent degrees of freedom per observable event. The phase structure of C encodes exactly these: magnitude r and phase θ in $z = r \cdot e^{i\theta}$. $U(1)$ is the automorphism group of this structure — the symmetry of the compiled scalar field. Gauge invariance is not imposed from outside; it is the symmetry group of the compiled amplitude algebra.

Full gauge group identification requires C₉ and C₁₀ (compilation of equivalence classes and gauge invariance) together with anomaly cancellation (T6). These remain candidate theorems. The present paper establishes $U(1)$ as a consequence of the Division Algebra Theorem.

§5 The Measurement Theorem

A direct consequence of the C₅ dual-channel structure is a compiled definition of measurement that removes a potential category error from the framework: a measurement residue is NOT a local violation of C₁.

§5.1 Channel Decomposition [C from C₅]

Definition (Channel Decomposition). A C₅ observable event decomposes the continuation state into two coupled channels: $s = (S, V)$, where S is the shape channel (the propagating continuation, satisfying C₁ at every step) and V is the value channel (the residue-preserving history, satisfying C₂ at every step). The channels are complementary projections of the same C₅ event: one moves, one remains.

The coupling between the channels is the wave equation. Shape channel carries the kinetic term $(\partial_t \phi)^2$ — what is moving. Value channel carries the gradient feedback term $c^2(\nabla \phi)^2$ — what the current value is relative to neighbors. The balance condition $\square \phi = 0$ is the equilibrium of the two-channel coupling. The compiled Lagrangian:

$$L = (\text{shape channel})^2 - c^2(\text{value channel})^2 = (\partial_t \phi)^2 - c^2(\nabla \phi)^2$$

reads architecturally as: shape channel drives the field forward in time; value channel resists by reporting spatial curvature back. The physical path is the one where these balance exactly.

§5.2 Facts as Temporary Fixed Points [C]

Lemma (Temporary Fixed Point). *A committed measurement residue is a temporary fixed point in the value projection but not in the full state.*

Proof. A committed residue satisfies $V_{t+1} = V_t$ in the value channel (the value is committed; C₂ preserves it). The shape channel satisfies $S_{t+1} \neq S_t$ (C₁: lawful continuation distinct from current state). Therefore the full state $s_{t+1} = (S_{t+1}, V_{t+1}) \neq (S_t, V_t) = s_t$. C₁ is satisfied globally. The measured dimension is locally stationary; the complete state is not. $\square$

Measurement Theorem [C]. *A C₅ event generates a stable observable fact by committing a value-channel residue into reachable history. Collapse is a C₂ history-update, not a C₁ transition.*

Proof. A C₅ event requires simultaneous propagation and residue. The propagation updates the shape channel: $S_t \rightarrow S_{t+1}$, satisfying C₁. The residue is, by definition, the component of the C₅ event that does not propagate — it remains at the current node when the shape channel moves. Therefore $V_t \rightarrow V_H$, where V_H enters the history graph. By C₂, reachable history cannot be deleted: $H_{t+1} \supseteq H_t$. Therefore V_H persists as a fact. By the Temporary Fixed Point Lemma, the full state continues. C₁ is not violated.

Fact = fixed residue in V (governed by C₂) + continued evolution in S (governed by C₁)

Collapse is a history update operating on V . It belongs to C_2 , not C_1 . This resolves the category error: a committed measurement is not a terminal state; it is a committed node in the reachable history graph that feeds into the shape channel as a boundary condition for subsequent evolution. $\square$

§5.3 Born Rule Route [C conditional on C_7]

The channel decomposition supplies the Hilbert space structure. The comparison functional on the value channel must satisfy four conditions compiled from the primitives:

(1) C-valued: T_2 requires all amplitudes to be C-valued. (2) C-linear in the second argument: linearity of C compiles from $C_1 + C_{1a}$. (3) Conjugate-linear in the first: C has conjugation; adjoint compiles from C's structure. (4) Positive definite: C_0 requires distinct residues to remain distinguishable; self-comparison of a nonzero residue must yield a positive real.

A functional satisfying (1)–(4) is a Hermitian positive-definite inner product. The Hilbert space structure of the value channel is compiled from $T_2 + C_0 + C_{1a}$, not assumed.

Remaining gap (Ω_B): Gleason's theorem requires Hilbert space dimension at least 3. The dimension of the state space comes from the spatial degrees of freedom (C_7 geometry). Once C_7 is compiled, Gleason applies: $\mu(P) = \langle \psi | P | \psi \rangle$ for all projection operators P . The Born rule is compiled, not assumed. C_7 is the single upstream gate.

§6 Action Emergence

§6.1 Discrete Path Cost

Definition 6.1 (Discrete path). A discrete path is an ordered sequence $\gamma = (s_0, s_1, \dots, s_n)$ where each step $s_i \rightarrow s_{i+1}$ is an admissible transition satisfying C_3 , L_3 , L_4 , and C_8 at every step.

Definition 6.2 (Path cost). The local transition cost c assigns a positive real cost to each admissible adjacent transition. The total path cost is the accumulated sum $C(\gamma) = \sum_i c(s_i, s_{i+1})$.

Three properties of c , each derived from the primitive chain:

(i) Positivity: $c(s, s') > 0$ for all admissible $s \neq s'$. A zero-cost transition is undetectable (C_0 violation). The identity transition at zero cost is excluded by C_1 .

(ii) **Locality:** $c(s, s')$ is defined only for L_4 -adjacent pairs. Non-adjacent transitions are compiled out by L_4 .

(iii) **Cost equals compiled Lagrangian density:** From C8 (Paper 1), the compiled free-field Lagrangian density is $L = (\partial_t \phi)^2 - c^2(\nabla \phi)^2$. The local transition cost is:

$$c(s_i, s_{i+1}) = L(\phi(s_i), \partial \phi(s_i)) \cdot \Delta t \cdot \Delta V$$

§6.2 Non-Circularity of the Derivation

A potential objection: does defining cost via L and then extremizing the integral of L merely recover what was put in?

The answer is no. The logical direction is:

$$\text{Primitive constraints} \rightarrow \text{local transition cost density } L \rightarrow \text{accumulated path cost } S \rightarrow \text{extremality condition } \delta S = 0$$

The Lagrangian L enters here as a previously compiled local transition cost density, not as a variational principle. This paper does not derive the form of L ; L is established in Paper 1 from Co–C7. The present theorem derives why the integral of this already-compiled local cost is extremized by admissible continuation paths. The compiled L and the action functional $S = \int L$ are different objects: L is the cost per step; S is the accumulated total. No circular dependency exists.

§6.3 Continuum Limit [C conditional on C7]

Theorem 6.3 [C conditional on C7]. *In the continuum limit of the admissible redistribution geometry (C7), the discrete path cost converges to the action functional.*

$$C(\gamma) = \sum_i L(\phi(s_i), \partial \phi(s_i)) \cdot \Delta t \cdot \Delta V \rightarrow \int L(\phi, \partial \phi) d^4x = S[\gamma]$$

Proof sketch. The C7 geometry (Paper 1) admits a continuum limit in which the graph Laplacian converges to the usual Laplacian operator. In this limit, the sum over discrete nodes becomes a Riemann integral. Field regularity — continuity of ϕ along admissible paths — follows from C2 (history preservation). The cost scaling is consistent with the compiled Lagrangian density by Definition 6.2(iii). $\square$

§6.4 C1-Consistency and the Transition Residual [C]

Definition 6.4 (Transition Residual). The transition residual at spacetime point (x, t) is the quantity:

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$$R(x,t) = \square\phi(x,t) = \partial_t^2\phi - c^2\nabla^2\phi$$

The residual measures the failure of local content redistribution to balance according to the compiled conservation law. $R = 0$ means local continuation balances exactly — the shape and value channels are in equilibrium. $R \neq 0$ means there is an uncompensated transition surplus at (x,t) , violating the compiled C_3/L_3 conservation at that point.

Definition 6.5 (C1-consistency). A field configuration $\phi(x,t)$ is C1-consistent at (x,t) if the transition residual vanishes: $R(x,t) = 0$.

Theorem 6.6 [C]. $\phi(x,t)$ is C1-consistent at every point if and only if $\square\phi = 0$ everywhere.

Proof. Immediate from Definition 6.5. The theorem makes the equivalence between the discrete notion (C1-consistent local continuation) and the analytic PDE (vanishing transition residual) explicit. The bridge is the compiled wave equation, which C8 (Paper 1) establishes as the unique admissible free-field evolution. $\square$

§6.5 The Extremal Theorem [C conditional on C7]

Theorem 6.7 [C conditional on C7] (Extremal Theorem). A field configuration $\phi(x,t)$ is C1-consistent at every point if and only if $\delta S[\phi] = 0$.

Proof (forward). Suppose ϕ is C1-consistent everywhere, i.e., $\square\phi = 0$ everywhere (Theorem 6.6). Compute the Euler-Lagrange equations for $L = (\partial_t\phi)^2 - c^2(\nabla\phi)^2$. Since L has no explicit ϕ dependence, $\partial L/\partial\phi = 0$. The remaining terms give $-\partial_t(2\partial_t\phi) + \partial_i(2c^2\partial_i\phi) = 0$, which is $-2\square\phi = 0$. The Euler-Lagrange equations are satisfied everywhere. By standard variational calculus, this is equivalent to $\delta S[\phi] = 0$ for all variations $\delta\phi$ vanishing at the boundary. $\square$

Proof (backward). Suppose $\delta S[\phi] = 0$ for all boundary-vanishing variations. By standard variational calculus, the Euler-Lagrange equations hold at every interior point: $\square\phi = 0$ everywhere. By Theorem 6.6, ϕ is C1-consistent at every point. $\square$

The equivalence chain in full:

$$C1\text{-consistent everywhere} \Leftrightarrow R(x,t) = 0 \Leftrightarrow \text{Euler-Lagrange equations} \Leftrightarrow \delta S = 0$$

Each equivalence is bidirectional. None requires imported physics. The compiled L from C8 produces the wave equation as its Euler-Lagrange output — the same equation C1-consistency requires.

Remark on extremality versus minimality. Physical trajectories extremize the action — $\delta S = 0$ includes minima, maxima, and saddle points. For the free field with $L = (\partial_t\phi)^2 - c^2(\nabla\phi)^2$, the

action is indefinite and physical solutions are saddle points of S . The proof above is stated correctly for extremality throughout. Describing Hamilton's principle as a minimum principle is accurate only for restricted problem classes and is avoided here.

§6.6 Hamilton's Principle as a Compiled Theorem [C conditional on C7]

Theorem 6.8 [C conditional on C7] (Hamilton's Principle). *For the free field satisfying Co–C5, the physical field configuration — the one satisfying the compiled transition rules at every point — is the admissible extremal configuration of $S[\varphi] = \int L \, d^4x$ with fixed boundary conditions.*

Proof. Physical configurations satisfy C1-consistency everywhere by definition (admissible under all compiled constraints). By Theorem 6.7, these are exactly the configurations satisfying $\delta S = 0$. $\square$

Hamilton's principle is not a rule imposed on nature. It is the global expression of a local requirement: at every point, the transition residual must vanish. The action accumulates the price of the path. When every local step pays exactly the compiled price and no more, the path is the admissible extremal.

Formulation	Status
	External postulate, assumed
	Compiled from Co–C5, conditional on C7

§7 The Independence Spine

Theorem [C] (Independence Spine). *The condition $[T_A, T_B] = 0$ for independent continuation operators is the algebraic expression of C1a. It appears at three abstraction levels of the compiled theory and produces three compiled results from the single axiom.*

Level 1: Transition Operators to Conservation [C]

If independent T_A creates net content and independent T_B compensates, then T_B 's behavior must depend on T_A 's output: $T_A = T_A(T_B)$. The commutator $[T_A, T_B] = 0$ blocks this coordination channel. Compiled result: $C_3 \rightarrow L_3$. Conservation is not imposed; it is the closure signature of independent composition.

Level 2: Scalar Algebra to C [C]

Scalar weights are continuation operations. $[a_A, a_B] = 0$ is required (Level 1 applied to the amplitude layer). H fails this — Case A of the simplicity argument shows every faithful representation of H contains non-commuting independent updates. C satisfies it automatically. Compiled result: C is the unique admissible scalar continuation algebra.

Level 3: Sector Symmetry to Gauge Factorization [C conditional on T₅, T₆]

Independent gauge sectors with generators $X_i \in G_1$ and $X_j \in G_2$ must satisfy $[X_i, X_j] = 0$. The gauge group factorizes: $G = G_1 \times G_2 \times \dots$. The scalar level contributes $U(1)$ from the phase structure of C. Full identification requires T₅ (sector commutativity) and T₆ (gauge group uniqueness via anomaly cancellation).

Level	Independent objects	C _{1a} compiled result
Transition algebra	State transitions T_A, T_B	No coordination channel => conservation [C]
Scalar algebra	Amplitude weights a_A, a_B	No order dependence => C forced [C]
Symmetry sectors	Gauge generators X_i, X_j	Sector commutation => G factorizes [C cond.]

One axiom. Three compiled results. Conservation, scalar field uniqueness, and gauge factorization are all instances of the same question: can independent distinctions evolve without hidden coordination? The commutator $[T_A, T_B]$ measures that coordination. When it vanishes, independence is genuine.

§8 Quantum Extension [T — Bridge, Not Result]

The following establishes the compiled structure of the path integral as a bridge to quantum mechanics. The classical Extremal Theorem is established; the path integral measure remains an open problem (Ω_C below).

Theorem 6.8 selects the single C₁-consistent path. The quantum theory generalizes: all C₁-admissible paths contribute, not just the admissible extremal:

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$$\langle \varphi_f | \varphi_i \rangle = \int_{\{C1\text{-admissible}\}} D[\varphi] \cdot \exp(iS[\varphi]/\hbar)$$

Component	Source	Status
All C1-admissible paths	C1: every state admits continuation	[C]
$\exp(iS/\hbar)$ weight	C scalar field from Division Algebra Theorem	[C]
$\hbar \rightarrow 0$ selects $\delta S=0$	Stationary phase recovers Theorem 6.8	[C conditional]
Measure $D[\phi]$	C7 lattice product measure	[T] — open

The path integral's structure is compiled. The measure construction — defining $D[\phi]$ rigorously on C1-admissible field configurations — is the remaining open problem. The C7 adjacency lattice provides a natural ultraviolet cutoff and a candidate product measure over lattice sites. Whether this extends to a renormalizable continuum QFT measure depends on the renormalization group structure, which is not yet compiled from Co–C5.

§9 Epistemic Status

Claim	Status	Remaining gap
Frobenius reduction	[C]	None
R elimination	[C]	None
H elimination (all representations)	[C]	Simplicity argument complete
Division Algebra Theorem (T2 promoted)	[C]	Promoted from [T] to [C]
U(1) from C phase structure	[C]	U(1) factor only; full group needs T6
Channel decomposition (S,V)	[C]	None

Claim	Status	Remaining gap
Measurement Theorem (collapse = C2 update)	[C]	None
Hilbert inner product from value channel	[C]	None (from T2 + Co + C1a)
C1-consistent $\Leftrightarrow \square\phi = 0$	[C]	None
Extremal Theorem ($\delta S=0 \Leftrightarrow$ C1-consistent)	[C*]	Inherits C7 gap
Hamilton's Principle (free field)	[C*]	Inherits C7 gap
Path integral structure (components)	[C]	Structural identification done
Path integral measure $D[\phi]$	[T]	Product measure / renormalization open
Born rule (T3)	[T]	Gleason needs C7 $\rightarrow \dim \geq 3$
CCR (T4)	[T]	Stone-von Neumann + representation selection
Gauge factorization (T5, T6)	[T]	Anomaly cancellation argument open
Interacting fields $V(\phi) \neq 0$	[T]	Requires C9 coupling compiled

[C*] = compiled conditional on C7 geometry uniqueness.

§10 Open Problems

Ω_1 — C7 Geometry Uniqueness (most urgent). The single upstream gap blocking the Extremal Theorem, Hamilton's Principle, Born rule (T3), and CCR (T4). Does maximal symmetry of the admissible redistribution geometry uniquely force flat space in the continuum limit? Closing C7 would simultaneously promote multiple conditional results to fully compiled and open the Born rule route through Gleason.

Ω_2 — Born Rule (Ω_B). Given the Hilbert structure compiled from T2 + Co + C1a in §5.3, Gleason's theorem applies once the state space dimension is at least 3. Dimension comes from

C7. The compiled result would be: $\mu(P) = \langle \psi | P | \psi \rangle$ for all projection operators P , derived rather than postulated.

Ω₃ — Gauge Group Uniqueness (T₅, T₆). The C_{1a} spine at Level 3 requires $[X_i, X_j] = 0$ for independent gauge sectors. Once sector commutativity is proved (T₅), the gauge group factorizes. Whether anomaly cancellation then uniquely selects $U(1) \times SU(2) \times SU(3)$ is T₆. Both require C₉ and C₁₀ as upstream inputs.

Ω₄ — Interacting Fields. The free-field Lagrangian has no potential term. Interactions require $V(\phi)$ compiled from C₉ (equivalence class coupling). Once V is compiled, the Extremal Theorem extends by the same proof structure. Paper 4 target.

Ω₅ — Path Integral Measure. The C₇ lattice provides a natural cutoff and candidate product measure. Whether this extends to a renormalizable continuum QFT measure depends on the renormalization group structure. Connects to T₆ and gauge structure.

§11 Program Summary

Paper	Central claim	Key input
Paper 1	Conservation and locality compiled from $\{C_0, C_1, C_{1a}, C_2, C_5\}$; L ₃ , L ₄ are not primitives	All five primitives
This paper (Paper 2)	C is unique scalar field [C]; Hamilton's principle compiled [C*]; Measurement Theorem [C]; Independence Spine [C]	C _{1a} + simplicity of H; C ₈ + C ₇
Paper 4 (planned)	Interacting fields; action with $V(\phi)$	C ₉ coupling

The primitive set $\{C_0, C_1, C_{1a}, C_2, C_5\}$ generates: conservation, locality, the scalar field, Hamilton's principle, and a compiled measurement theory — without importing a single physical law from outside.

Appendix A — Topological Consequences of C₁

C₁ requires that every admissible state has a successor state distinct from itself. Equivalently, the transition vector field on the state manifold has no zeros. By the Poincaré-Hopf theorem, a

vector field on a compact manifold without zeros exists if and only if the Euler characteristic of the manifold satisfies $\chi(M) = 0$. Therefore the state space of any C_1 -admissible compact system lives on a manifold with vanishing Euler characteristic. Examples: even-dimensional tori $T^{\{2n\}}$, odd-dimensional spheres $S^{\{2n-1\}}$.

This is a topological corollary of C_1 requiring no additional input. It connects the discrete primitive C_1 to a classical topological constraint on admissible state manifolds.

Appendix B — Relation to Quaternionic Quantum Mechanics

Quaternionic quantum mechanics (QQM) is a well-developed mathematical framework and the H elimination theorem does not dispute its internal consistency. The distinction is in what is being eliminated.

In QQM, the state space is a right-H-module: vectors are complex and scalars act on the right, but the inner product takes values in H. The algebra of observables is the set of H-linear self-adjoint operators. This framework is internally consistent.

The Division Algebra Theorem eliminates H as the scalar continuation algebra under a specific operational interpretation: every nonzero element of D acts as an admissible independent amplitude update on independent subsystems. Under this interpretation, independent phase updates must commute (C_{1a}). H fails this in every faithful representation (Section 4.3.2).

In QQM, independent amplitude updates do NOT compose by simple quaternionic multiplication across independent subsystems. The right-module structure requires conventions about ordering — specifically, which way quaternions act and on which side. That additional structure is precisely what C_{1a} prohibits: it introduces an ordering convention between independent operations, unlicensed under C_0 – C_5 .

The theorem is therefore narrower than "quaternions are impossible in physics" and exactly as strong as "H-valued scalar continuation algebra violates C_{1a} under any faithful representation." QQM and the Division Algebra Theorem are compatible because they operate at different levels of the theory.

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